

Unideals Custom Knowledge Base

For a Google Gemini AI chatbot supporting the Campus-Mart / Unideals student marketplace

This document is designed as a comprehensive source of truth for chatbot retrieval, prompt grounding, support answers, API-aware help, and user-facing platform explanations.

Project	Unideals / Campus-Mart
Product type	Full-stack peer-to-peer student marketplace
Main users	Verified university students, buyers, sellers, admins, support staff
Frontend	React 18, Vite, Tailwind CSS, React Router
Backend	Express 5, MongoDB, Mongoose, JWT auth
Generated for	Google Gemini AI custom knowledge base

How To Use This PDF As A Knowledge Base

- Upload or index this PDF as a reference document for Gemini, Vertex AI Search, a vector database, or any retrieval-augmented chatbot layer.
- Chunk by headings and question-answer pairs. Recommended chunk size: 500-900 tokens with 80-120 token overlap.
- Treat this as authoritative for platform behavior, but prefer live API/database data for product availability, prices, user account state, and real-time messages.
- For user-specific answers, the chatbot should ask the application backend for authenticated user details instead of guessing.

Platform Summary

- Unideals is a campus-focused marketplace where students can buy, sell, discover, wishlist, report, and boost second-hand or campus-relevant items.
- The platform emphasizes trusted campus communities, email verification, public meetup safety, clear listings, and moderation tools.
- Users can register with email/password or Google OAuth, verify email, manage profile/address details, list products with images, search/filter products, save favorites, contact support, and use chat assistance.
- Admins and support users have protected tools to list users, list products, change user status, moderate product status, soft-delete, or hard-delete products.

Knowledge The Chatbot Should Contain

Knowledge area	Details to answer
Marketplace purpose	What Unideals is, who it is for, why campus students use it.
Accounts	Registration, login, Google OAuth, verification, password reset, blocked accounts, deletion.
Listings	How to sell, listing requirements, categories, images, prices, conditions, drafts, unlisting, relisting, deletion.
Buying	Browse, search, filters, product detail pages, wishlist, contacting sellers, safety guidance.
Payments	Cash, UPI, or both; platform does not hold escrow or funds.
Boosting	Monthly limits, durations, active boost limits, and why boosting improves visibility.
Reports and safety	How to report products/users, reasons, response expectations, safe meetup advice.
Support	Contact support, report issue, suggest feature, expected response window, support email shown in UI.
Technical	Frontend routes, backend APIs, environment variables, security controls, data models.
Chatbot behavior	Answer style, guardrails, fallback responses, escalation rules, Gemini integration notes.

Primary User Roles

Role	Capabilities
Guest	Browse public product feed, search, view product details, read terms/privacy, visit login/signup.
Verified/registered user	Create listings, wishlist products, manage profile and addresses, report products/users, contact support, boost eligible listings.

Role	Capabilities
Seller	Create product listings or drafts, manage listed products, unlist, relist, soft-delete, boost products, respond to buyers.
Buyer	Search/browse products, save wishlist items, inspect product details, contact seller, follow safety guidance.
Support	Access admin-protected support routes if role is support; review users/products depending on backend permissions.
Admin	Admin auth, list users/products, update statuses, soft-delete/hard-delete products.

Frontend Route Knowledge

Route	Access	Purpose
/	Public	Home / product feed.
/search	Public	Search results page.
/product/:id	Public	Product detail page.
/category/:categoryName	Public	Category browser, also used for boosted-products views.
/price	Public	Price range filter or pricing model view.
/termscondition	Public/user layout	Terms and conditions / legal document view.
/privacy-policy	Public	Privacy policy page.
/login	Public	Sign in.
/signup	Public	Create account.
/forgot-password	Public	Request password reset email.
/reset-password/:token	Public	Set a new password after token validation.
/verify-email	Public	Verify email with OTP/token flow.
/checkEmail	Public	Post-signup email confirmation prompt.
/profile	Protected	Profile overview.
/settings	Protected	Account settings.
/subscription	Protected	Subscription plan UI.
/wishlist	Protected	Saved products.
/myorders	Protected	Order/deal history UI.
/chat	Protected	Messaging UI and assistant access.
/notification	Protected	Activity notifications UI.
/upload	Protected	Multi-step product listing flow.
/productlisted	Protected	Post-listing confirmation.
/contact	Protected	Help, support, reports, feature suggestions.

Product Categories And Options

Value	Label	Typical items
electronics	Electronics	Laptops, tablets, calculators, headphones, chargers, phones.
study_material	Study Material	Textbooks, notes, guides, question banks.
hostel_essentials	Hostel Essentials	Mattresses, buckets, lamps, room supplies.
clothing	Clothing	T-shirts, hoodies, jackets, ethnic wear.
accessories	Accessories	Watches, bags, wallets, earphones.
lab_equipment	Lab Equipment	Lab coats, kits, instruments, records.
sports	Sports	Cricket bats, footballs, rackets, sports gear.
fitness	Fitness	Dumbbells, resistance bands, gym equipment.
vehicles	Vehicles	Bicycles, scooters, bike accessories.
event_passes	Event Passes	Concerts, fests, workshops, competitions.
others	Others	Anything that does not fit the above categories.

Product Listing Requirements

- Basic information: title is required and must be at least 3 characters; description is required and must be at least 10 characters.
- Category and condition are required for listed products.
- Usage duration and purchase date are required by the frontend listing flow.
- Images: at least 1 image and maximum 3 images for a listed product. Image URLs must be valid HTTP/HTTPS URLs.
- Pricing: selling price and original price are required in frontend validation; selling price cannot exceed original price.
- Payment preference: cash, UPI, or both.
- Pickup address: required before final listing submission; includes address line, city, state, pincode, and optional mobile/additional info.
- Draft products may omit fields that are required for a fully listed product.

Product Status And Lifecycle

Status	Meaning
draft	Incomplete or saved listing not yet published.
listed	Visible active product listing.
sold	Product has been sold or marked sold.
unlisted	Seller removed listing from active browsing without deleting it.
blocked	Listing is restricted by admin/moderation.
soft deleted	Product has is_deleted=true; hidden from normal browsing and may be cleaned up later.

User Account Knowledge

- Users have name, email, password, avatar, mobile, gender, subscription tier, role, status, email verification state, addresses, refresh token, location fields, and wishlist references.
- Roles are user, admin, and support.

- User statuses are active, inactive, and suspended.
- Email verification is required for trust and may affect full participation. Unverified accounts have a TTL index that can remove them after 24 hours.
- Password reset uses reset tokens and expiry values.
- Google OAuth is supported through /api/auth/google, /api/auth/google/callback, and /api/auth/google/exchange.
- Account deletion is permanent from the user's perspective and removes account access/listings according to controller behavior.

Backend API Reference - Mounted Routes

Area	Method	Endpoint	Auth	Purpose
Health	GET	/health	Public	Server health check with status, uptime, timestamp.
Auth	POST	/api/auth/register	Public	Create account.
Auth	POST	/api/auth/login	Public	Local login.
Auth	GET	/api/auth/google	Public	Start Google OAuth.
Auth	GET	/api/auth/google/callback	Public	Handle Google callback.
Auth	POST	/api/auth/google/exchange	Public	Exchange OAuth code for app session.
Auth	POST	/api/auth/verify-email	Public	Verify email.
Auth	GET	/api/auth/check-verification	Public	Check email verification status.
Auth	POST	/api/auth/resend-verification	Public	Resend verification.
Auth	POST	/api/auth/forgot-password	Public	Request password reset.
Auth	GET	/api/auth/reset-password/:token	Public	Validate reset token before submit.
Auth	POST	/api/auth/reset-password/:token	Public	Set new password.
Auth	POST	/api/auth/refresh-token	Cookie	Rotate/refresh access token.
Auth	GET	/api/auth/logoutUser	Public	Clear cookies and invalidate refresh token.
User	GET	/api/user/userProfile	User	Get authenticated user profile.
User	PUT	/api/user/updateProfile	User	Update profile.
User	DELETE	/api/user/deleteAccount	User	Delete account.
Product	GET	/api/product	Public	List products with pagination/search/filter/sort.
Product	GET	/api/product/boosted	Public	List boosted products.
Product	GET	/api/product/search	Public	Full text product search with q.
Product	GET	/api/product/search-suggestions	Public	Autocomplete suggestions; q min 2 chars.
Product	GET	/api/product/:id	Public	Get product by id and increment views.
Product	POST	/api/product	User	Create listing or draft; rate limited to 5/min.
Product	DELETE	/api/product/:id	User	Soft-delete owned listing.

Area	Method	Endpoint	Auth	Purpose
Product	PATCH	/api/product/:id/unlist	User	Unlist owned product.
Product	PATCH	/api/product/:id/relist	User	Relist owned product.
Product	GET	/api/product/user/my-products	User	Get current user's products.
Product	GET	/api/product/user/drafts	User	Get current user's drafts.
Wishlist	GET	/api/wishlist	User	Get wishlist.
Wishlist	POST	/api/wishlist/add	User	Add product to wishlist.
Wishlist	POST	/api/wishlist/remove	User	Remove product from wishlist.
Wishlist	POST	/api/wishlist/toggle	User	Toggle wishlist state.
Wishlist	GET	/api/wishlist/check/:productId	User	Check whether product is saved.
Address	POST	/api/address	User	Create address.
Address	GET	/api/address	User	List user's addresses.
Address	GET	/api/address/:addressId	User	Get address by id.
Address	PUT	/api/address/:addressId	User	Update address.
Address	DELETE	/api/address/:addressId	User	Delete address.
Address	PATCH	/api/address/:addressId/default	User	Set default address.
Report	POST	/api/report/product/:productId	User	Report a product.
Report	POST	/api/report/user/:userId	User	Report a user.
Boost	GET	/api/boost/me/summary	User	Get boost usage summary.
Boost	POST	/api/boost/products/:productId	User	Boost listing.
ImageKit	GET	/api/imagekit/auth	User	Get signed upload parameters.
Chat	POST	/api/chat/assistant	Public/rate limited	Assistant message endpoint; 30/min.
Admin	POST	/api/admin/auth/login	Public	Admin login.
Admin	POST	/api/admin/auth/refresh-token	Public	Refresh admin session.
Admin	GET	/api/admin/auth/me	Admin/support	Current admin/support user.
Admin	POST	/api/admin/auth/logout	Admin/support	Admin logout.
Admin	GET	/api/admin/users	Admin/support	List users.
Admin	PATCH	/api/admin/users/:id/status	Admin/support	Update user status.
Admin	GET	/api/admin/products	Admin/support	List products for moderation.
Admin	PATCH	/api/admin/products/:id/status	Admin/support	Update product status.
Admin	PATCH	/api/admin/products/:id/soft-delete	Admin/support	Soft-delete product.
Admin	DELETE	/api/admin/products/:id	Admin/support	Hard-delete product.

Important Note About Unmounted Backend Modules

The repository contains route/model/service files for conversations, deals, messages, notifications, reviews, subscriptions, and support tickets. In the inspected app.js, only auth, user, product, report, address, imagekit, wishlist, admin, boost, and chat are mounted. A chatbot should not claim unmounted endpoints are available unless app.js is

updated.

Boosting Rules

Tier	Monthly boost limit	Duration	Max active boosts
Base User	2	1 hour	1
Pro	10	3 hours	1
Pro Plus	30	3 hours	3

Report Reasons

- Spam or advertisement
- Fake or misleading listing
- Inappropriate content
- Fraud or scam
- Prohibited item
- Duplicate listing
- Wrong category
- Sold or unavailable
- Other

Security And Privacy Knowledge

- Backend uses Helmet with crossOriginResourcePolicy disabled for media compatibility.
- CORS allows configured FRONTEND_URL, ADMIN_FRONTEND_URL, localhost/127.0.0.1 ports 5173/5174, and one LAN origin currently in code.
- Requests are sanitized for NoSQL injection by deleting keys containing '\$' or '.'.
- String body and param values are sanitized using xss.
- JSON payload limit is 10kb.
- Cookies are supported through cookie-parser and CORS credentials.
- Axios client uses VITE_API_BASE_URL and withCredentials=true.
- Auth interceptor refreshes access tokens on 401, stores accessToken in localStorage, clears auth on refresh failure, and dispatches an account-blocked browser event on 403 accountBlocked responses.
- Privacy policy states that Unideals does not sell, rent, or trade personal data.
- Users can request access, correction, deletion, portability, and objection by emailing privacy@campusmart.in.

Terms, Safety, And Dispute Guidance

- Unideals is a technology marketplace and is not a party to user transactions.
- The platform does not hold funds, take title to goods, or guarantee quality, safety, or legality of listings.
- Users should meet in public, well-lit campus locations and inspect items before paying.
- Payments/refunds/returns are agreed directly between buyer and seller. There is no platform-level returns system.

- Prohibited activities include illegal goods, counterfeit products, weapons, drugs, alcohol, prescription medication, fraud, harassment, fake reviews, scraping, hacking, commercial reselling without permission, and impersonation.
- Disputes with Unideals follow Indian law. Terms mention courts in Vellore, Tamil Nadu.
- Users must be 18+ and currently enrolled at a recognized Indian university or college according to the terms document.

Support And Contact Knowledge

- Help & Support page has three actions: Contact support, Report an issue, Suggest a feature.
- Displayed support response time: under 2 hours during campus hours, 9 AM - 9 PM.
- Displayed support email: hi@unideals.in.
- Reports for immediate safety emergencies should direct users to campus security first; UniDeals reports are reviewed within 24 hours according to the UI copy.
- Contact support form topics include Account assistance, Deal inquiries, and Other.
- Report issue categories include Fraud or Scam Seller, Safety Concern, App Bug / Glitch, Harassment or abuse, Spam or Bot account, and Inappropriate Content.

Current Assistant Endpoint Behavior

Intent	How it responds
product_search	Infers category and budget, searches active listed products, returns up to matching products and suggestions.
recommendations	Returns active listings sorted by boost, views, and recency if matching recommendations exist.
price_estimate	Uses median of recent active listings in inferred category to suggest low/fair/high price range.
listing_assistant	Creates starter title, category, description, and tags.
faq	Answers safety, refund/payment, listing, boost, and wishlist questions from built-in FAQ.
conversation	Explains that it can search, estimate, write listings, answer marketplace questions, and recommend deals.

Recommended Gemini Chatbot System Instruction

You are Unideals Assistant, a helpful chatbot for a campus student marketplace. Use the knowledge base to answer questions about accounts, listings, buying, selling, safety, support, policies, categories, boost plans, and app navigation. Be concise but complete. If the user asks about live product availability, prices, account state, order state, or messages, use backend/API data when available or explain that live data is needed. Never invent product inventory, payment guarantees, refund promises, or legal commitments. For emergencies or immediate physical safety concerns, tell the user to contact campus security or local authorities first, then report in Unideals.

Chatbot Guardrails

- Do not promise refunds, returns, escrow, buyer protection, or platform payment holding unless implemented later.
- Do not expose secrets, environment variables, private keys, refresh tokens, or internal tokens.
- Do not provide legal, financial, or safety guarantees. Give practical guidance and escalate where needed.
- Do not say a product is available unless confirmed by live API data.
- When user asks how to delete account, warn that deletion is permanent.
- When user reports fraud or safety risks, ask for listing/user details and direct them to the report flow. For immediate danger, direct them to campus security first.

- When answer depends on logged-in identity, ask the app/backend for authenticated user data.
- When an endpoint is unmounted, describe it as code present in repository but not exposed by current app.js.

Chatbot Question-Answer Seed Set

Q: What is Unideals?

A: Unideals is a peer-to-peer student marketplace for buying, selling, and discovering products within a campus community.

Q: Who can use Unideals?

A: The intended users are university/college students, with terms requiring users to be 18+ and enrolled at a recognized Indian university or college.

Q: How do I create an account?

A: Go to /signup, enter your details, then verify your email. Users can also sign in through Google OAuth if configured.

Q: How do I verify my email?

A: Use the email verification flow at /verify-email or the post-signup /checkEmail page. You can request another verification email through resend verification.

Q: How do I reset my password?

A: Open /forgot-password, submit your email, then use the reset link/token at /reset-password/:token to set a new password.

Q: How do I sell something?

A: Go to /upload, add basic details, choose category and condition, upload 1-3 images, set price/payment preference, choose pickup address, accept terms, then submit.

Q: How many product photos can I upload?

A: A listed product needs at least one image and supports a maximum of three images.

Q: Can I save a draft?

A: Yes. The backend supports product drafts, and /api/product/user/drafts returns the current user's drafts.

Q: Can I edit a listing?

A: The frontend has an updateProduct API helper, but the inspected backend mounted product routes do not include PATCH /api/product/:id for general editing. Unlist/relist/delete are mounted.

Q: How do I remove my listing?

A: You can soft-delete your owned product with DELETE /api/product/:id or unlist it with PATCH /api/product/:id/unlist.

Q: How do I relist an item?

A: Use the relist action, which calls PATCH /api/product/:id/relist for owned products.

Q: What categories are available?

A: Electronics, Study Material, Hostel Essentials, Clothing, Accessories, Lab Equipment, Sports, Fitness, Vehicles, Event Passes, and Others.

Q: What payment methods are supported?

A: Product listings can mark payment preference as cash, UPI, or both. The platform does not hold payments.

Q: Is there a return policy?

A: There is no platform-level returns system. Buyers should inspect items before completing a deal and agree terms directly with the seller.

Q: How do I report a scam?

A: Use the report option on the product/user or contact support. Report reasons include fraud or scam, fake/misleading listing, inappropriate content, prohibited item, and other.

Q: How quickly are reports reviewed?

A: The support UI says safety reports are reviewed within 24 hours. Immediate safety emergencies should go to campus security first.

Q: How does wishlist work?

A: Logged-in users can add, remove, toggle, and check products in their wishlist through /api/wishlist endpoints.

Q: How does boosting work?

A: Boosting makes a listing more visible. Limits depend on subscription tier: Base User 2/month for 1 hour, Pro 10/month for 3 hours, Pro Plus 30/month for 3 hours and up to 3 active boosts.

Q: How do I contact support?

A: Open /contact. You can contact support, report an issue, or suggest a feature. The displayed support email is hi@unideals.in.

Q: What if my account is blocked?

A: The frontend clears stored auth and displays an account restricted message when the backend returns accountBlocked. Contact support for help.

Q: What data does Unideals collect?

A: Privacy policy lists account data, verification data, listing data, usage data, and in-app messages.

Q: Does Unideals sell my data?

A: The privacy policy says Unideals does not sell, rent, or trade personal data.

Q: Can I delete my account?

A: Yes, through settings/profile or DELETE /api/user/deleteAccount. The UI warns this is permanent and cannot be undone.

Q: What should I do before meeting a seller?

A: Meet in a public, well-lit campus area, inspect the item, avoid sharing home addresses, and pay only after verification.

Q: What can the chatbot help with?

A: It can search products, recommend listings, estimate fair prices, draft listing descriptions, explain platform features, and answer support/policy questions.

Data Model Summary For Technical Answers

Model	Important fields
User	name, email, password, avatar, mobile, gender, subscription, role, status, is_email_verified, tokens, address_details, subscription_details, current_lat/current_long, wishlist.

Model	Important fields
Product	seller_id, title, slug, description, category, condition, attributes, images, original_price, selling_price, is_negotiable, payment_preference, pickup_address_snapshot, location, status, boost fields, views_count, is_deleted.
Address	user, line1, line2, city, state, pincode, isDefault.
Wishlist	Stores user/product wishlist relation; user-facing API also uses User wishlist references.
Boost	product_id, user_id, tier, starts_at, expires_at, duration_hours, status.
Report	reporter_id, target_id, target_model, reason, description, status, admin_note, reviewed_at.
Conversation	product_id, seller_id, buyer_id, last message preview, product snapshot, unread count, active/delete flags.
Message	conversation_id, sender_id, receiver_id, message_type, text, deal_id, message_status, delivery/read/edit/delete metadata.
Deal	conversation_id, original/current price, offer history, offer/status lifecycle, pickup, payment confirmation, item received, completion/cancellation.

Environment And Setup Knowledge

Area	Variables / commands
Backend env	PORT, NODE_ENV, FRONTEND_URL, ADMIN_FRONTEND_URL, MONGO_URL, SECRET_KEY_ACCESS_TOKEN, SECRET_KEY_REFRESH_TOKEN, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, GOOGLE_REDIRECT_URI, IMAGEKIT_PUBLIC_KEY, IMAGEKIT_PRIVATE_KEY, IMAGEKIT_URL_ENDPOINT, RESEND_API_KEY.
Frontend env	VITE_API_BASE_URL, VITE_IMAGEKIT_PUBLIC_KEY, VITE_IMAGEKIT_URL_ENDPOINT, VITE_EMAILJS_SERVICE_ID, VITE_EMAILJS_TEMPLATE_ID, VITE_EMAILJS_PUBLIC_KEY, VITE_SUPPORT_EMAIL.
Backend dev	cd backend; npm install; npm run dev; default port 5000.
Frontend dev	cd frontend; npm install; npm run dev; default port 5173.
Health check	GET http://localhost:5000/health.

Recommended Retrieval Metadata Tags

- account, auth, login, signup, verification, password-reset, google-oauth
- product, listing, selling, draft, images, pricing, category, condition
- buying, search, wishlist, product-detail, filters
- boost, subscription, base-user, pro, pro-plus
- report, safety, scam, fraud, support, contact
- privacy, terms, data, deletion, legal
- api, backend, frontend, routes, environment, setup
- assistant, gemini, prompt, guardrails, faq

Final Implementation Notes For Gemini

- Best architecture: Gemini model + retrieval layer over this PDF + live backend tools for user/product state.
- For product search, call existing /api/product/search or a custom product lookup tool instead of relying only on static PDF knowledge.

- For assistant replies, the existing `/api/chat/assistant` can be upgraded to call Gemini and include retrieved PDF chunks in the prompt.
- Store conversation history carefully. Do not include passwords, tokens, private keys, or sensitive documents in prompts.
- Add citations internally by section name so support/admin users can audit the chatbot's source.